

OOPS CONCEPT

FUNCTIONS:

- A [Java](#) function, also known as a method, is a block of code designed to perform a specific task, defined by the syntax ``public static returnType methodName(parameters) { ... }``.

Types of functions:

- Java supports several types of functions, each with its own specific purpose. These include static functions, instance functions, recursive functions, and lambda functions.

Return keyword:

- The return keyword in Java is used to exit from a method and optionally pass back a value to the method caller. It serves as a control flow statement that terminates the execution of the method in which it appears.

Components of class in java:

- The Component class is the superclass of all components.
public void add(Component c):
public void setSize(int width, int height):
public void setLayout(LayoutManager lm):
public void setVisible(boolean status):

Types of Variables and its uses:

- It defines variables as containers that hold data values and notes there are three types: local, instance, and static.

What is Constructor:

- In Java, a constructor is a special method used to initialize objects.

Constructor advantages and its types:

- A parameterised constructor allows passing a list of parameters in the constructor definition. Therefore, it gives an advantage to set the starting values of the object. Just like a method can have parameters of different data types, in the same way, a constructor can also have a parameter list of multiple data types.

Ways to initialize the Object:

- An object in Java can be initialized in different ways:
- **Using new keyword** – Most common method.

```
Student s = new Student();
```

- **Using a constructor** – Initialize values when creating the object.

```
Student s = new Student("Arun");
```

- **Using a method** – Set values through a method.

```
s.setName("Arun");
```

“this” keyword:

- **this Keyword**

this refers to the **current object**.

Main uses:

- To differentiate instance variables and local variables.
- To call the current class constructor.
- To call current class methods.

Example:

```
class Student {  
    String name;  
  
    Student(String name) {  
        this.name = name;  
    }  
}
```

Here, this.name refers to the **instance variable**.

Static Block & Instance Block:

- A **static block** is executed **only once when the class is loaded**.

```
class Demo {  
    static {  
        System.out.println("Static Block");  
    }  
}
```

Use: To initialize static variables or perform one-time initialization.

Instance Block

An **instance block** is executed **every time an object is created**, before the constructor.

```
class Demo {  
    {  
        System.out.println("Instance Block");  
    }  
}
```

Use: To initialize instance variables or perform common initialization for objects.